

# CityVision — Running the App with Docker

## Urban Scene Semantic Segmentation

The application runs as **two containers**: a **backend** (FastAPI prediction API) and a **frontend** (Streamlit UI that consumes the API). Both are started together with Docker Compose.

### Prerequisites

- **Docker Desktop** installed and *running* (the whale icon in the system tray is steady; the app shows “Engine running”).
- A terminal opened at the project root:  
C:\Users\DFILATOVA\Desktop\CityVision

Check that Docker is running:

```
docker version
```

If a **Server:** section is printed, the engine is up. If you only see the **Client:** block plus an error, start Docker Desktop first.

### Step 1 — Start the containers

From the project root:

```
docker compose up -d
```

- **-d** runs them in the background (detached).
- The *first* run builds the images and can take a few minutes (it downloads PyTorch). Later runs are fast.
- If you changed the code, rebuild first: `docker compose up --build -d`

### Step 2 — Check they are running

```
docker compose ps
```

You should see **two** containers, both **running**:

```
cityvision-backend-1  running  0.0.0.0:8000->8000/tcp
cityvision-frontend-1 running  0.0.0.0:8501->8501/tcp
```

## Step 3 — Open the app

- **Application (frontend):** <http://localhost:8501>
- **API documentation (backend):** <http://localhost:8000/docs>

On the very first load, wait ~10 seconds for the backend to load the model, then refresh if you see a connection error.

## Step 4 — Use the app

1. In the sidebar, select an **image** from the dropdown (e.g. `frankfurt_000000_000294`).
2. Click **Predict mask**.
3. The three panels appear: **real image**, **real mask**, and **predicted mask**, plus the detected-class breakdown.

## Step 5 — Stop the containers

```
docker compose down
```

This stops and removes both containers. The built images stay cached, so the next start is quick.

## Quick reference

| Goal                         | Command                                   |
|------------------------------|-------------------------------------------|
| Start (images already built) | <code>docker compose up -d</code>         |
| Start after code changes     | <code>docker compose up --build -d</code> |
| Check status                 | <code>docker compose ps</code>            |
| View logs                    | <code>docker compose logs -f</code>       |
| Stop                         | <code>docker compose down</code>          |

## Troubleshooting

`error during connect ...dockerDesktopLinuxEngine`

Docker Desktop's engine is not running. Open Docker Desktop and wait for "Engine running", then retry.

`az / other command "not recognized"`

Not related to running the app; only relevant for cloud deployment.

**Frontend shows "API not reachable"**

The backend is still starting (model load). Wait a few seconds and refresh. Confirm with `docker compose ps` that `cityvision-backend-1` is running.

**Port already in use**

Another process holds port 8501 or 8000. Stop it, or change the port mapping in `docker-compose.yml`.